SHRIHARI M V

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Professional Summary

Fresher software engineer with hands-on experience building full-stack, backend, and AI-powered applications through independent projects using Python, React, FastAPI, REST APIs, and modern AI tools. Built end-to-end applications involving authenticated frontend workflows, API services, stateful agent systems, retrieval pipelines, testing, validation, telemetry, and deployment. Strong foundation in data structures, object-oriented programming, SQL, APIs, Git, and debugging, with the ability to learn unfamiliar technologies and build, test, and improve software independently.

Technical Skills

Languages	Python, Java, SQL, C, TypeScript
Software Engineering	Data Structures & Algorithms, Object-Oriented Programming, REST APIs, API Integration, Async Programming, Git, GitHub, Pytest
Frontend	React, React Router, TypeScript, TanStack Query, Zod
Backend	FastAPI, Pydantic, Swagger/OpenAPI, JWT/Bearer Authentication
AI Engineering	LangChain, LangGraph, RAG, Tool Calling, Embeddings, Semantic Search, Vector Databases, ChromaDB, Ollama
ML & Data	Scikit-learn, XGBoost, PyTorch, Feature Engineering, Cross Validation, Model Evaluation, Pandas, NumPy
Engineering & Deployment	Docker, MLflow, Weights & Biases, Structured Logging, Telemetry, ONNX Runtime

Projects

AttriShield – Employee Attrition Decision-Support Platform GitHub

React • TypeScript • FastAPI • Python • XGBoost • MLflow • JWT

- Built an end-to-end employee attrition decision-support platform connecting a **React/TypeScript frontend** with **FastAPI REST services** for prediction, batch analysis, and analytics workflows.
- Designed reusable frontend components with **protected routes, JWT/Bearer authentication, controlled forms, and centralized API communication**, integrating client workflows with backend inference services.
- Implemented **Zod response validation and TanStack Query** for server-state management, caching, asynchronous request handling, and mutation workflows, with explicit loading, success, and error states.
- Integrated **XGBoost inference, SHAP-based explanations, MLflow experiment tracking, and application telemetry** to provide interpretable and observable prediction workflows.

StudyMentor – Personal Learning & Retention Coach GitHub

Next.js • React • TypeScript • FastAPI • MongoDB • Pydantic • LangGraph

- Built a backend-focused adaptive learning platform with **FastAPI, asynchronous application services, MongoDB, and deterministic learning services** for questions, reviews, scheduling, analytics, and learner memory.
- Designed a stateful **LangGraph workflow** that combines structured learner state with LLM-based activity selection, tool calling, conditional routing, checkpointing, and persistent learner memory while keeping database and business logic outside the agent.
- Engineered reliability around LLM workflows using **Pydantic structured outputs, deterministic quality checks, bounded retries, asynchronous execution, and human-review escalation** for failed generations.
- Validated the backend with **pytest (19 tests passing)** and integration experiments covering tool calling, learner-memory loading, memory-to-planner behavior, LangGraph execution, and deterministic quality checks.

CarnaticPrep – Retrieval-Augmented Knowledge System GitHub

Python • ChromaDB • Sentence Transformers • CrossEncoder • Ollama • FastAPI • RAGAS

- Designed a modular hybrid RAG pipeline combining **dense retrieval, BM25, adaptive query routing, weighted Reciprocal Rank Fusion (RRF), and CrossEncoder reranking** for domain-specific knowledge retrieval.
- Built corpus-aware indexing pipelines using **structure-aware chunking, schema-aware metadata normalization, and adaptive PDF/CSV processing** for heterogeneous knowledge sources.
- Developed an end-to-end **FastAPI RAG system** supporting vector indexing, configurable retrieval workflows, structured logging, Swagger/OpenAPI documentation, and grounded response generation using Ollama.

Education

B.E. Computer Science & Engineering (Data Science) 2022 – 2026
Bangalore Institute of Technology CGPA: **8.6/10**

PCMB 2020 – 2022
Sri Kumarans PU College Percentage: **94.83%**

Experience

MindMatrixEd – Generative AI Intern Feb 2026 – May 2026

- Explored AI-assisted Android application workflows using Google AI Studio, Android Studio and Kotlin contributing to feature prototyping and user-centric product design.